Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The ordered pair (λ, μ) such that the points $(\lambda, \mu, -6)$, $(3, 2, -4)$ and $(9, 8, -10)$ become collinear is

(A) $(3, 4)$

(B) $(5, 4)$

(C) $(4, 5)$

(D) $(4, 3)$

Q.2

The line $\frac{x-1}{1} = \frac{y-2}{-2} = \frac{z-1}{3}$ and the plane $x + 2y + z = 6$
meet in

(A) no point

(B) only one point

(C) infinitely many points

(D) None of these

Q.3

Let the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + \beta = 0$. Then (α, β) equals

(A) $(-6, 7)$

(B) $(5, -15)$

(C) $(-5, 5)$

(D) $(6, -17)$

Q.4

Equation of line in the plane $\pi \equiv 2\mathrm{x} - \mathrm{y} + \mathrm{z} - 4 = 0$ which is perpendicular to the line ℓ whose equation is
$\frac{x-2}{1} = \frac{y-2}{-1} = \frac{z-3}{-2}$ and which passes through the point of intersection of ℓ and π is -

(A) $\frac{x-2}{3} = \frac{y-1}{5} = \frac{z-1}{-1}$

(B) $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$

(C) $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z+1}{1}$

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(D) $\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-1}{1}$

Q.5

The line passing through the points $(5, 1, a)$ and $(3, b, 1)$ crosses the yz-plane at the point $\left(0, \frac{17}{2}, \frac{-13}{2}\right)$. Then

(A) $a = 2, b = 8$

(B) $a = 4, b = 6$

(C) $a = 6, b = 4$

(D) $a = 8, b = 2$

Q.6

Given the line $L : \frac{x-1}{3} = \frac{y+1}{2} = \frac{z-3}{-1}$ and the plane $\pi : \mathrm{x} - 2\mathrm{y} = \mathrm{z}$. Of the following assertions, the only one that is always true is

(A) L is $\perp$ to π

(B) Llies in π

(C) L is paralel to π

(D) None of these

Q.7

The distance of point $A(-2, 3, 1)$ from the line PQ through $P(-3, 5, 2)$, which makes equal angles with the axes is

(A) $2/\sqrt{3}$

(B) $\sqrt{14/3}$

(C) $16/\sqrt{3}$

(D) $5/\sqrt{3}$

Q.8

If the straight line $\frac{x-x_0}{\ell} = \frac{y-y_0}{m} = \frac{z-z_0}{n}$ is parallel to the plane $ax + by + cz + d = 0$ then which one of the following is correct?

(A) $\ell + \mathrm{m} + \mathrm{n} = 0$

(B) $a + b + c = 0$

(C) $\frac{\mathrm{a}}{\ell} + \frac{\mathrm{b}}{\mathrm{m}} + \frac{\mathrm{c}}{\mathrm{n}} = 0$

(D) $\mathrm{a}\ell + \mathrm{bm} + \mathrm{cn} = 0$

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Q.9

The distance between the line $\vec{r} \cdot 2\hat{i} - 2\hat{j} + 3\hat{k}$ $+\lambda(\hat{i} - \hat{j} + 4\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - 5\hat{j} + \hat{k}) = 5$ is

(A) $\frac{10}{3\sqrt{3}}$

(B) $\frac{10}{9}$

(C) $\frac{10}{3}$

(D) $\frac{3}{10}$

Q.10

The angle between the line $\frac{x-2}{a} = \frac{y-2}{b}$ $= \frac{z-2}{c}$ and the plane $ax + by + cz + 6 = 0$ is

(A) $\sin^{-1}\left(\frac{1}{\sqrt{a^2+b^2+c^2}}\right)$

(B) 45°

(C) 60°

(D) 90°

Q.11

A variable plane which remains at a constant distance $3\mathrm{p}$ from the origin cut the coordinate axes at A, B and C. The locus of the centroid of triangle ABC is

(A) $x^{-1} + y^{-1} + z^{-1} = p^{-1}$

(B) $x^{-2} + y^{-2} + z^{-2} = p^{-2}$

(C) $x + y + z = p$

(D) $x^2 + y^2 + z^2 = p^2$

Q.12

The distance of the point $(1, -2, 3)$ from the plane $\mathrm{x} - \mathrm{y} + \mathrm{z} = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$ is

(A) 1

(B) 2

(C) 4

(D) $2\sqrt{3}$

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Q.13

The angle between the pair of planes represented by equation $2x^2 - 2y^2 + 4z^2 + 6xz + 2yz + 3xy = 0$ is

(A) $\cos^{-1}\left(\frac{1}{3}\right)$

(B) $\cos^{-1}\left(\frac{4}{21}\right)$

(C) $\cos^{-1}\left(\frac{4}{9}\right)$

(D) $\cos^{-1}\left(\frac{7}{\sqrt{84}}\right)$

Q.14

The equation of the plane which passes through the line of intersection of planes $\vec{r} \cdot \overrightarrow{n_1} = q_1, \vec{r} \cdot \overrightarrow{n_2} = q_2$ and is parallel to the line of intersection of planes $\vec{r} \cdot \overrightarrow{n_3} = q_3$ and $\vec{r} \cdot \overrightarrow{n_4} = q_4$ is

(A) $\left[\overrightarrow{n_2}\overrightarrow{n_3}\overrightarrow{n_4}\right]\left(\vec{r} \cdot \overrightarrow{n_1} - q_1\right) = \left[\overrightarrow{n_1}\overrightarrow{n_3}\overrightarrow{n_4}\right]\left(\vec{r} \cdot \overrightarrow{n_2} - q_2\right)$

(B) $\left[\overrightarrow{n_1}\overrightarrow{n_2}\overrightarrow{n_3}\right]\left(\vec{r} \cdot \overrightarrow{n_4}q_4\right) = \left[\overrightarrow{n_4}\overrightarrow{n_3}\overrightarrow{n_1}\right]\left(\vec{r} \cdot \overrightarrow{n_2} - q_2\right)$

(C) $\left[\overrightarrow{n_4}\overrightarrow{n_3}\overrightarrow{n_1}\right]\left(\vec{r} \cdot \overrightarrow{n_4} - q_4\right) = \left[\overrightarrow{n_1}\overrightarrow{n_2}\overrightarrow{n_3}\right]\left(\vec{r} \cdot \overrightarrow{n_2} - q_2\right)$

(D) None of these

Q.15

Under which one of the following conditions will the two planes $x + y + z = 7$ and $\alpha x + \beta y + \gamma z = 3$, be parallel (but not coincident)?

(A) $\alpha = \beta = \gamma = 1$ only

(B) $\alpha = \beta = \gamma = \frac{3}{7}$ only

(C) $\alpha = \beta = \gamma$

(D) None of the above

Q.16

The image of the point $(-1, 3, 4)$ in the plane $x - 2y = 0$ is

(A) $(8, 4, 4)$

(B) $\left(-\frac{17}{3}, -\frac{19}{3}, 4\right)$

(C) $(15, 11, 4)$

(D) $\left(-\frac{4}{5}, -\frac{13}{5}, 4\right)$

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Q.17

Two system of rectangular axes have the same origin. If a plane cuts them at distance a, b, c and a, b, c from origin then

(A) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}-\frac{1}{a'^2}+\frac{1}{b'^2}-\frac{1}{c'^2}=0$

(B) $\frac{1}{a^2}-\frac{1}{b^2}-\frac{1}{c^2}+\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}=0$

(C) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}-\left(\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}\right)=0$

(D) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}+\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}=0$

Q.18

A tetrahedron has vertices at $O(0,0,0), A(1,2,1), B(2,1,3)$ and $C(-1,1,2)$. Then the angle between the faces OAB and ABC will be

(A) $\cos^{-1}\left(\frac{19}{35}\right)$

(B) $\cos^{-1}\left(\frac{17}{31}\right)$

(C) 30°

(D) 90°

Q.19

Equation of the plane through three points A, B, C with position vectors $-6\hat{i}+3\hat{j}+2\hat{k}, 3\hat{\hat{i}}-2\hat{j}+4\hat{k}, 5\hat{1}+7\hat{j}+3\hat{k}$ is

(A) $\hat{\mathrm{r}}(\hat{\mathrm{i}}-\hat{\mathbf{j}}+7\hat{\mathbf{k}})+23=\mathbf{0}$

(B) $\hat{\mathrm{r}}(\hat{i}+\hat{j}+7\hat{k})+23=0$

(C) $\hat{r}(\hat{i}+\hat{j}-7\hat{k})+23=0$

(D) $\hat{\mathrm{r}}(\hat{\mathrm{i}}-\hat{\mathbf{j}}-\mathbf{7}\hat{\mathbf{k}})+\mathbf{23}=\mathbf{0}$

Q.20

The equation of the plane through the intersection of the planes $\mathrm{x}+2\mathrm{y}+3\mathrm{z}-4=0, 4\mathrm{x}+3\mathrm{y}+2\mathrm{z}+1=0$ and passing through the origin will be

(A) $x+y+z=0$

(B) $17x+14y+11z=0$

(C) $7x+4y+z=0$

(D) $17x+14y+z=0$

Answers & Solutions

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Q.1 Answer:

$(5, 4)$

Solution:

If the given points $(\lambda, \mu, -6), (3, 2, -4)$ and $(9, 8, -10)$ are collinear then $\frac{\lambda-3}{9-3} = \frac{\mu-2}{8-2} = \frac{-6+4}{-10+4} \Rightarrow \lambda = 5, \mu = 4$

Q.2 Answer:

infinitely many points

Solution:

D.R. of given line are $1, -2, 3$ and the d.r. of normal to the given plane are $1, 2, 1$. Since $1 \times 1 + (-2) \times 2 + 3 \times 1 = 0$, therefore, the line is parallel to the plane, Also, the base point of the line $(1, 2, 1)$ lies in the given plane. $(1 + 2 \times 2 + 1 = 6$ is true $)$

Hence, the given line lies in the given plane. Alternatively, any point on the given line is $(t + 1, -2t + 2, 3t + 1)$

It lies in the given plane $x + 2y + z = 6$ if $t + 1 + 2(-2t + 2) + 3t + 1 = 6$

i.e. if $0t = 0$, which is true for all real t. Hence every point on the given line lies in the given plane i.e. the line lies in the plane.

Q.3 Answer:

$(-6, 7)$

Solution:

$\because$ The line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + \beta = 0$

$\therefore$ Point $(2, 1, -2)$ lies on the plane i.e. $2 + 3 + 2\alpha + \beta = 0$

or $2\alpha + \beta + 5 = 0$ ...(i)

Also normal to plane will be perpendicular to line, $\therefore 3 \times 1 - 5 \times 3 + 2 \times (-\alpha) = 0 \Rightarrow \alpha = -6$

From equation (i) we have, $\beta = 7$

$\therefore (\alpha, \beta) = (-6, 7)$

Q.4 Answer:

$\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$

Solution:

Let direction ratios of the line be $\langle a, b, c \rangle$, then $2a - b + c = 0$

$a - b - 2c = 0$

i.e., $\frac{a}{3} = \frac{b}{5} = \frac{c}{-1}$

$\therefore$ direction ratios of the line are $(3, 5, -1)$

Any point on the given line is $(2 + \lambda, 2 - \lambda, 3 - 2\lambda)$. It lies on the plane π if $2(2 + \lambda) - (2 - \lambda) + (3 - 2\lambda) = 4$

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i.e., $4+2\lambda-2+\lambda+3-2\lambda=4$; i.e., $\lambda=-1$

$\therefore$ the point of intersection of the line and the plane is $(1,3,5)$

$\therefore$ equation of the required line is $\frac{x-1}{3}=\frac{y-3}{5}=\frac{z-5}{-1}$

Q.5 Answer:

$a=6, b=4$

Solution:

Equation of line through $(5,1,a)$ and $(3,b,1)$ is $\frac{x-5}{-2}=\frac{y-1}{b-1}=\frac{z-a}{1-a}=\lambda$

$\therefore$ Any point on this line is a $[-2\lambda+5,(b-1)\lambda+1,(1-a)\lambda+a]$

It crosses yz plane where $-2\lambda+5=0$

$\lambda=\frac{5}{2}$

$\therefore \left(0,(b-1)\frac{5}{2}+1,(1-a)\frac{5}{2}+a\right)=\left(0,\frac{17}{2},\frac{-13}{2}\right)$

$\Rightarrow (b-1)\frac{5}{2}+1=\frac{17}{2}$ and $\quad(1-a)\frac{5}{2}+a=-\frac{13}{2}$

$\Rightarrow b=4$ and $a=6$

Q.6 Answer:

Llies in π

Solution:

Since $3(1)+2(-2)+(-1)(-1)=3-4+1=0$

$\therefore$ given line is $\perp$ to the normal tothe plane i.e., given line is parallel to the given plane. Also $(1,-1,3)$ lies on the plane $x-2y-z=0$ if $1-2(-1)-3=0$ i.e. $1+2-3=0$ which is true $\therefore$ Llies in plane π.

Q.7 Answer:

$\sqrt{14/3}$

Solution:

Here, $\alpha=\beta=\gamma$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1 \quad \therefore \cos\alpha=\frac{1}{\sqrt{3}}$

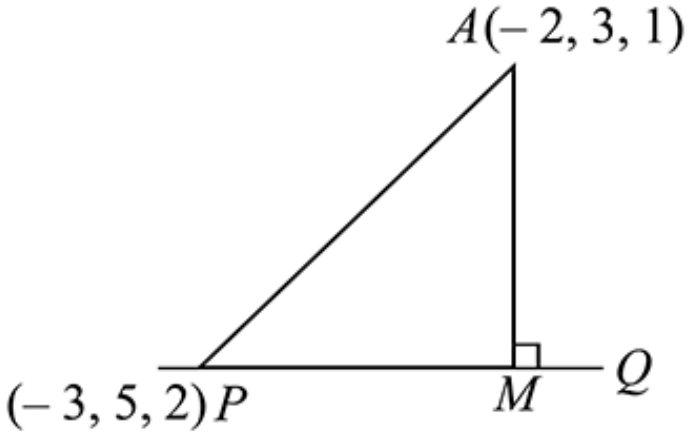
Direction cosines of PQ are $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

$PM=$ Projection of AP on $PQ=\left|(-2+3)\frac{1}{\sqrt{3}}+(3-5)\frac{1}{\sqrt{3}}+(1-2)\frac{1}{\sqrt{3}}\right|=\frac{2}{\sqrt{3}}$

and $AP=\sqrt{(-2+3)^2+(3-5)^2+(1-2)^2}=\sqrt{6}$

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$AM = \sqrt{(AP)^2 - (PM)^2} = \sqrt{6 - \frac{4}{3}} = \sqrt{\frac{14}{3}}$



Q.8 Answer:

$a\ell + bm + cn = 0$

Solution:

If the line is parallel to the plane then $a\ell + bm + cn = 0$

Q.9 Answer:

$\frac{10}{3\sqrt{3}}$

Solution:

It is obvious that the given line and plane are parallel. Given point on the line is $A(2, -2, 3)$.

$B(0, 0, 5)$ **is a point on the plane. Therefore,** $\overrightarrow{AB} = (2-0)\hat{i} + (-2-0)\hat{j} + (3-5)\hat{k}$

Then distance of B **from the plane = projection of**

$\overrightarrow{AB}$ **on vector** $\hat{i} + 5\hat{j} + \hat{k}$

$P = \left| \frac{(2\hat{i} - 2\hat{j} - 2\hat{k}) \cdot (\hat{i} + 5\hat{j} + \hat{k})}{\sqrt{1+25+1}} \right| = \left| \frac{2-10-2}{\sqrt{27}} \right| = \frac{10}{3\sqrt{3}}$

Q.10 Answer:

90°

Solution:

Obviously the line perpendicular to the

plane because $\frac{a}{a} = \frac{b}{b} = \frac{c}{c}$ **i.e., their direction ratios are proportional.**

Q.11 Answer:

$x^{-2} + y^{-2} + z^{-2} = p^{-2}$

Solution:

Let equation of the variable plane be

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

This meets the coordinate axes at $\mathrm{A}(\mathrm{a}, 0, 0), \mathrm{B}(0,\ \mathrm{b}, 0)$ **and** $\mathrm{C}(0, 0, \mathrm{c})$

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Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the centroid of the $\triangle \mathrm{ABC}$. Then

$\alpha=\frac{\mathrm{a}+0+0}{3}, \beta=\frac{0+\mathrm{b}+0}{3}, \gamma=\frac{0+0+\mathrm{c}}{3}$

$\therefore \quad \mathrm{a}=3\alpha, \mathrm{b}=3\beta, \mathrm{c}=3\gamma$

Plane (1) is at constant distance $3\mathrm{p}$ from the origin, so

$3p=\frac{\left|\frac{0}{a}+\frac{0}{b}+\frac{0}{c}-1\right|}{\sqrt{\left(\frac{1}{a}\right)^2+\left(\frac{1}{b}\right)^2+\left(\frac{1}{c}\right)^2}}$

$\Rightarrow \frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{9p^2} \quad \ldots(3)$

From (2) and (3), we get

$\frac{1}{9\alpha^2}+\frac{1}{9\beta^2}+\frac{1}{9\gamma^2}=\frac{1}{9p^2}$

$\Rightarrow \alpha^{-2}+\beta^{-2}+\gamma^{-2}=\mathrm{p}^{-2}$

Generalizing α, β, γ, locus of centroid $\mathrm{P}(\alpha, \beta, \gamma)$ is $x^{-2}+y^{-2}+z^{-2}=p^{-2}$

Q.12 Answer:

1

Solution:

Equation of the line through $(1,-2,3)$ parallel

to the line $\frac{\mathrm{x}}{2}=\frac{\mathrm{y}}{3}=\frac{\mathrm{z}-1}{-6}$ is

$\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=r$ (say)

Then any point on (1) is $(2r+1, 3r-2, -6r+3)$ If this point lies on the plane $x-y+z=5$ then

$(2r+1)-(3r-2)+(-6r+3)=5 \Rightarrow r=\frac{1}{7}$

Hence the point is $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$

Distance between $(1,-2,3)$ and $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$

$=\sqrt{\left(\frac{4}{49}+\frac{9}{49}+\frac{36}{49}\right)}=\sqrt{\left(\frac{49}{49}\right)}=1$

Q.13 Answer:

$\cos^{-1}\left(\frac{4}{9}\right)$

Solution:

$2x^2-2y^2+4z^2+(6xz+2yz+3xy=0$

or $2x^2+x(6z+3y)-2y^2+4z^2+2yz=0$

$x=\frac{-(6z+3y)\pm\sqrt{36z^2+9y^2+36yz-8(-2y^2+4z^2+2yz)}}{4}$

$x=\frac{-(6z+3y)\pm\sqrt{(2z+5y)^2}}{4}$

$\Rightarrow x=\frac{-(6z+3y)\pm(2z+5y)}{4}$

or $\quad 2x-y+2z=0, x+2y+2z=0$

$\therefore \quad$ Angle between planes is $\cos^{-1}\left(\frac{4}{9}\right)$.

Q.14 Answer:

$\left[\overrightarrow{n_2} \overrightarrow{n_3} \overrightarrow{n_4}\right]\left(\vec{r} \cdot \overrightarrow{n_1}-q_1\right)=\left[\overrightarrow{n_1} \overrightarrow{n_3} \overrightarrow{n_4}\right]\left(\vec{r} \cdot \overrightarrow{n_2}-q_2\right)$

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Solution:

$\left(\vec{r} \cdot \overrightarrow{n_1} + \lambda \vec{r} \cdot \overrightarrow{n_2} = q_1 + \lambda q_2\right.$

where λ is a parameter.

So, $\overrightarrow{n_1} + \lambda \overrightarrow{n_2}$ is normal to plane (i). Now, any plane

parallel to the line of intersection of the planes

$\vec{r} \cdot \overrightarrow{n_3} = q_3$ **and** $\vec{r} \cdot \overrightarrow{n_4} = q_4$ **is ofthe form** $\vec{r} \cdot \left(\overrightarrow{n_3} \times \overrightarrow{n_4}\right) = d$

Hence, we must have $\left[\overrightarrow{n_1} + \lambda \overrightarrow{n_2}\right] \cdot \left[\overrightarrow{n_3} \times \overrightarrow{n_4}\right] = 0$

or $\left[\overrightarrow{n_1 n_3 n_4}\right] + \lambda \left[\overrightarrow{n_2 n_3 n_4}\right] = 0$ **or** $\lambda = \frac{-\left[\overrightarrow{n_1 n_3 n_4}\right]}{\left[\overrightarrow{n_2 n_3 n_4}\right]}$

on putting this value in Eq. (i), we have the equation of the required plane as

$\vec{r} \cdot \vec{n}_1 - q_1 = \frac{[\vec{n}_1 \vec{n}_3 \vec{n}_4]}{[\vec{n}_2 \vec{n}_3 \vec{n}_4]} (r \cdot \vec{n}_2 - q_2)$

or $\left[\overrightarrow{n_2 n_3 n_4}\right] \left(\vec{r} \cdot \overrightarrow{n_1} - q_1\right) = \left[\overrightarrow{n_1 n_3 n_4}\right] \left(\vec{r} \cdot \overrightarrow{n_2} - q_2\right)$

Q.15 Answer:

$\alpha = \beta = \gamma$

Solution:

Given equation of planes are:

$x + y + z = 7$ **and** $\alpha x + \beta y + \gamma z = 3$

For these planes to be parallel, coefficients of x, y **and** z **should be same i.e.** $\Rightarrow \quad \alpha = \beta = \gamma$

Q.16 Answer:

$\left(-\frac{4}{5}, -\frac{13}{5}, 4\right)$

Solution:

We know that the image $(\mathrm{x}, \mathrm{y}, \mathrm{z})$ **of point** $\left(\mathrm{x}_1, \mathrm{y}_1 \mathrm{z}_1\right)$

In a plane $ax + by + cz + d = 0$ **is**

$\frac{\mathrm{x}-\mathrm{x}_1}{\mathrm{a}} = \frac{\mathrm{y}-\mathrm{y}_1}{\mathrm{b}} = \frac{\mathrm{z}-\mathrm{z}_1}{\mathrm{c}} = \frac{-2(a\mathrm{x}_1+\mathrm{by}_1+\mathrm{cz}_1+\mathrm{d})}{\mathrm{a}^2+\mathrm{b}^2+\mathrm{c}^2}$

Thus, the image of point $(-1, 3, 4)$ **in a plane** $x - 2y = 0$ **is**

$\frac{\mathrm{x}+1}{1} = \frac{\mathrm{y}-3}{-2} = \frac{\mathrm{z}-4}{0} = \frac{-2[1(-1)+(-2)3+0(4)]}{1+0+4}$

$\frac{x+1}{1} = \frac{y-3}{-2} = \frac{z-4}{0} = \frac{-2(-7)}{5}$

$\mathrm{x} = \frac{14}{5} - 1 = \frac{9}{5}; \mathrm{y} = \frac{-28}{5} + 3 = \frac{-13}{5} \& \mathrm{z} = 4$

$\therefore$ **Image of** $(-1, 3, 4)$ **is** $\left(\frac{9}{5}, \frac{13}{5}, 4\right)$

Q.17 Answer:

$\frac{1}{\mathrm{a}^2} + \frac{1}{\mathrm{b}^2} + \frac{1}{\mathrm{c}^2} - \left(\frac{1}{\mathrm{a}'^2} + \frac{1}{\mathrm{b}'^2} + \frac{1}{\mathrm{c}'^2}\right) = 0$

Solution:

Let $\mathrm{a}, \mathrm{b}, \mathrm{c}$ **be the intercepts when** $\mathrm{Ox}, \mathrm{Oy}, \mathrm{Oz}$ **are taken as axes; then the equation of the plane is**

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

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Also, let a', b', c' be the intercepts when OX, OY, OZ are taken as axes; then in this case equation of the same plane is

$\frac{X}{a'} + \frac{Y}{b'} + \frac{Z}{c'} = 1 \quad \ldots(2)$

Now,(1) and (2) are equations of the same plane and in both the cases the origin is same.

Hence, length of the perpendicular drawn form the origin to the plane in both the case must be the same,

i.e., $\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}}}$

$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$

Q.18 Answer:

$\cos^{-1}\left(\frac{19}{35}\right)$

Solution:

Vector perpendicular to face OAB

$= OA \times OB = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = 5i - j - 3k \ldots(1)$

Vector perpendicular to face ABC

$= AB \times AC = \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ -2 & -1 & 1 \end{vmatrix} = i - 5j - 3k \quad \ldots(2)$

Since the angle between the faces = angle between their normals thus

$\cos\theta = \frac{5+5+9}{\sqrt{35}\sqrt{35}} = \frac{19}{35} \Rightarrow \theta = \cos^{-1}\left(\frac{19}{35}\right)$

Q.19 Answer:

$\hat{r}(\hat{i} - \hat{j} + 7\hat{k}) + 23 = 0$

Solution:

Equation of the plane passing through three points A, B, C with position vectors a, b, c is

$r(a \times b + b \times c + c \times a) = ab \times c$

So that if a, b, c represent the given vectors, the $(a \times b + b \times c + c \times a) =$

$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -6 & 3 & 2 \\ 3 & -2 & 4 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 4 \\ 5 & 7 & 3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 7 & 3 \\ -6 & 3 & 2 \end{vmatrix}$

$= i(12 + 4 - 6 - 28 + 14 - 9) - \hat{j}(-24 - 6 + 9 - 20 + 10 + 18)$

$+\hat{k}(12 - 9 + 21 + 10 + 15 + 42) = -13\hat{i} + 13\hat{j} - 91\hat{k}$

And $ab \times c = \begin{vmatrix} -6 & 3 & 2 \\ 3 & -2 & 4 \\ 5 & 7 & 3 \end{vmatrix} = 299$

So the required equation of the plane is $r \cdot (-13\hat{i} + 13\hat{j} - 91\hat{k}) = 299$

Or $\mathbf{r} \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) = -23$ **or** $r.\hat{\mathbf{i}} - \hat{\mathbf{j}} + 7\hat{\mathbf{k}}) + 23 = 0$

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Q.20 Answer:

$17x + 14y + 11z = 0$

Solution:

Equation of plane passing through intersecting the planes $x + 2y + 3z - 4 = 0$ **and** $4x + 3y + 2z + 1 = 0is, x + 2y3z - 4 + \lambda 04x + 3y + 2z + 1) = 0$

Since it is passing through origin, so $\lambda = 4$**.**

Hence the required equation is $17\mathrm{x} + 14\mathrm{y} + 11\mathrm{z} = 0.8$

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